

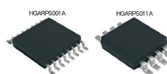
112G PAM4 connectivity. It is suitable for applications including enterprise Ethernet switches, security appliances, cloud interconnect routers, and optical transport systems.

Microchip

<https://www.microchip.com>

Magnetic angle sensors

Alps Alpine HGARPS001A and HGARPS011A are new magnetic angle sensors for power steering applications. They come in a single-die and a dual-die model. Both models achieve redundancy by



providing two output signals from a single product. These sensors provide resistance to outside magnetic disturbance, making the sensing more accurate, which is essential for the safety of the vehicle. They are meant for non-contact switches, linear position/angle, and rotation speed/direction detection. The sensors come with AEC-Q100 automotive reliability standards, making them suitable for use in harsh automotive environments.

Alps Alpine

<https://www.alpsalpine.com>

64-channel ultrasound scanner

STMicroelectronics' STHVUP64 is a highly integrated 64-channel ultrasound transmitter. It enables designers to enhance the image quality and shrink the form factor of handheld scanners. It can drive up to 256 probe elements directly, eliminating expensive, high-voltage switches usually needed between the transmitter and probe. The transmitter adds extra flexibility and a digital beam steering feature, thus enabling



extra space-saving for the designers. It offers multiple built-in features, such as noise blocking, thermal protection, under-voltage protection, and recirculating current protection. It can help designers reduce the BoM and save time while designing.

STMicroelectronics

<https://www.st.com>

Diodes for automotive applications

Nexperia's new Clip-bonded FlatPower (CFP) packaged diodes are optimised for low forward voltage (VF) and de-



liver low conduction losses, making them suitable for industrial and automotive applications. The new launch includes 32 planar

Schottky diodes and eight hyperfast recovery rectifiers housed in CFP15B package and has an operational range between 30-100V and 3-15A. The diodes meet AEC-Q101 automotive qualification standards, making them safe for use in harsh automotive environments. The smaller package of these diodes enables design engineers to save up to 60% of board space while maintaining a performance similar to the large and bulky diodes available in DPAK or SMB/C packages. The diodes can be used in advance driver assistance systems, electric vehicles, LED lighting, or electronic control unit applications relying on advanced high-density designs.

Nexperia

www.nexperia.com

EMBEDDED

AI vision microprocessors

Renesas RZ/V series of microprocessors enables AI processing of image

data from multiple cameras. These come with a built-in vision AI accelerator for accurate image recognition and multi-camera image support.



The microprocessors are equipped with two 64-bit Arm Cortex-A53 cores, which enables them to deliver high computing performance with a maximum operating frequency of 1GHz. The RZ/V2MA features a proprietary low-power dynamically reconfigurable processor accelerator, which can process vision AI at one tera operations per second, per watt. These features bring highly accurate image recognition capabilities for machine vision products, such as AI-equipped gateways, video servers, security gates, POS terminals, and robotic arms.

Renesas

<https://www.renesas.com>

Processors for embedded systems

AMD Ryzen Embedded V3000 series of processors meet the need of high-performing embedded systems. The processors feature a high-performance Zen 3 core to the V-Series portfolio.



Compared to the previous generation, the new processors offer up to 124% greater CPU performance, 50% improved memory transfer rate, 2X CPU core count, and improved I/O connectivity. The new processor family enables system designers to utilise a single-board design for a wide range of system configurations. The embedded processors can power a variety of diverse use cases ranging from virtual hyper-converged infrastructure to advanced systems at the edge.

AMD

<https://www.amd.com>



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